

Albert Cao

✉ albertcao@college.harvard.edu ◦ ☎ (301) 356-1522 ◦ 🌐 abeot ◦ in abeot ◦ 🌐 abeot.github.io

Education

Harvard University

Aug 2026 – May 2030 (expected)

Intended joint concentration in Mathematics and Computer Science

Cambridge, MA

- **Relevant Coursework:** Math 55a (Studies in Algebra and Group Theory), Stat 110 (Introduction to Probability), CS 1210 (Introduction to Theoretical Computer Science)
- **Activities:** AI Safety Student Team, Harvard Financial Analysts Club, Harvard-MIT Math Tournament, Startups @ Harvard, Harvard Computer Society, Poker Club

Experience

Intern

January 2026 – May 2026

Axilon (Startup)

New York, NY

- Built an agentic RAG system that generates Ignition SCADA projects from natural-language prompts
- Improved orchestration/harnesses, developed the deployment web interface, and wrote self-evaluation capabilities

Research Intern

March 2023 – January 2026

University of Michigan

Hybrid

- Developed deep learning models for drug toxicity prediction, incorporating previously neglected factors in drug-target interaction with molecular and biological features

Intern

June 2024 – May 2025

Johns Hopkins Applied Physics Laboratory

Laurel, MD

- Developed SMT-based control-flow graph recovery software, writing Go implementation for indirect jump resolution
- Built a low-level vulnerability analysis framework using reverse engineering tools with custom and proprietary LLMs

Publications

AI and machine learning for predicting in vitro/preclinical drug toxicities and adverse events in clinical patients.

Acta Pharmaceutica Sinica B, 2026 (co-author)

Machine Learning Model Predicts Clinical Adverse Events of Small Molecule Kinase Inhibitors in Cancer Patients Using On-/Off-Target Engagement and Tissue Selectivity. *Clinical Pharmacology & Therapeutics*, 2026 (co-author)

Machine Learning Prediction of On/Off Target-driven Clinical Adverse Events. *Pharmaceutical Research*, 2024 (first-author)

Academic Competition Awards

USA Math Olympiad Silver Medalist

2026

- Placed among the top 40 students on the USA Mathematical Olympiad, the nation's premier math olympiad

Regeneron Science Talent Search Scholar

2026

- Placed among the top 300 scholars in the 85th Regeneron Science Talent Search

AMC, AIME and USAMO Participant

2025

- Scored a 150/150 on the AMC 12, 12/15 on the AIME, and 21/42 on USAMO

USA Junior Math Olympiad Honors

2024

- Placed among the top 50 students on the USA Junior Mathematical Olympiad, scoring 23/42

USA Computing Olympiad, Gold Division

2023

Extracurriculars

Division I Fencer

2017 – Present

Harvard Fencing Team / DC Fencers Club

- A-rated fencer, top 50 nationally in Cadet Men's Épée, top 100 nationally in Junior Men's Épée
- Competing in NCAA Division I as a member of Harvard's fencing team.

Technical Skills

Languages: Python, Java, C++, Julia, TypeScript, C, Assembly

ML / AI: PyTorch, scikit-learn, agentic pipelines, RAG ◦ **Web:** React, Electron, REST APIs

Security: Ghidra, SMT solvers, control-flow analysis ◦ **Tools:** Lean 4, $\LaTeX$, Git